

SEVEN DISJOINT ALL-6-ISOLATING SETS NEED NOT EXIST IN A TREE

ABSTRACT. Boyer, Farrell, and Goddard proved that, for $k \leq 5$, every tree of order different from k has a $(k + 1)^*$ -coloring into all- k -isolating classes, with one exceptional tree, and left $k = 6$ open. We show that the corresponding seven-set statement fails. The two eight-vertex spiders with arm lengths 3, 2, 2 and 2, 2, 2, 1 have no seven pairwise disjoint nonempty all-6-isolating sets, even without an independence requirement. Among trees of order different from 6, they are the minimum-order trees with no 7^* -coloring into all-6-isolating classes: orders 1–5 and 7 all admit one, and order 6 is excluded. On the other hand, every tree has a 6^* -coloring whose color classes are all-6-isolating, so six is the optimal universal guarantee.

For a graph G and $A \subseteq V(G)$, let $N[A]$ be the closed neighborhood of A . The set A is *all- k -isolating* if every component of $G - N[A]$ has order less than k . Following Boyer, Farrell, and Goddard [1], an ℓ^* -coloring is a weak partition of $V(G)$ into ℓ independent color classes, where empty classes are allowed. They proved the $(k + 1)^*$ -coloring result for $k \leq 5$, constructed obstructions for $k \geq 7$, and left the case $k = 6$ open [1, Theorem 7 and the paragraph following Lemma 8]. We answer that case negatively. Since [1] states no result for $k = 6$, no statement there is contradicted; what is new below is the eight-vertex obstruction to seven all-6-isolating classes, whereas the positive 6^* -coloring guarantee is deduced from their Theorem 7 together with two exceptional cases.

Write $S_{r_1, \dots, r_d}$ for the spider whose arms have lengths $r_1, \dots, r_d$.

Theorem 1. *Every tree has a 6^* -coloring in which every color class is all-6-isolating. However, neither $S_{3,2,2}$ nor $S_{2,2,2,1}$ has seven pairwise disjoint nonempty all-6-isolating sets. Up to isomorphism, they are the smallest trees of order different from 6 admitting no 7^* -coloring all of whose color classes are all-6-isolating; every tree of order at most 7 other than those of order 6 admits one. In particular, neither spider has an ℓ^* -coloring with all color classes all-6-isolating for any $\ell \geq 7$; thus six is the optimal universal number of color classes.*

Proof. By [1, Theorem 7], every tree other than the subdivided claw O_7 and trees of order 5 has a 6^* -coloring whose color classes are all-5-isolating, and hence all-6-isolating. If a tree has order 5, every subset, including the empty set, is all-6-isolating, so any proper 6^* -coloring works. For O_7 , give two leaves one color and give the other five vertices distinct colors. This

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is a proper coloring with six nonempty classes, and each class has a closed neighborhood containing at least two vertices; hence at most five vertices remain. Thus every tree has the stated 6^* -coloring.

It remains to determine the smallest obstruction to seven sets. Let T be a tree of order 8, and call a leaf *bad* if its neighbor has degree 2. A singleton $\{x\}$ is not all-6-isolating if and only if x is a bad leaf. Indeed, if $d_T(x) \geq 2$, then $|N[x]| \geq 3$, so at most five vertices remain. If x is a leaf with neighbor y , then $T - N[x] = T - \{x, y\}$ has six vertices; it is connected exactly when $d_T(y) = 2$.

Suppose first that T has seven pairwise disjoint nonempty all-6-isolating sets, and let s of them be singletons. Since each nonsingleton has at least two vertices,

$$8 \geq s + 2(7 - s) = 14 - s,$$

so $s \geq 6$. Therefore a tree with at least three bad leaves cannot have seven such sets. Conversely, if T has at most two bad leaves, place all of them in one independent two-vertex set, adding a nonneighbor if there is only one and choosing any two distinct leaves if there are none. Use the remaining six vertices as singleton sets. The two-vertex set is all-6-isolating because its closed neighborhood has at least three vertices, and every singleton used is all-6-isolating. Hence an order-eight tree has seven pairwise disjoint nonempty all-6-isolating sets if and only if it has at most two bad leaves.

Now suppose that T has at least three bad leaves x_1, x_2, x_3 , with respective degree-two neighbors y_1, y_2, y_3 . These six vertices are distinct. Let p, q be the remaining vertices. The second neighbor of each y_i lies in $\{p, q\}$: it cannot be another y_j , since then $x_i y_i y_j x_j$ would be a component. Connectedness forces $pq \in E(T)$. Up to interchanging p and q , the three vertices y_i attach either all to p or in a 2–1 split. These two possibilities are precisely $S_{2,2,2,1}$ and $S_{3,2,2}$. Each has three bad leaves and therefore admits no seven such sets, even without independence. Since the empty set is not all-6-isolating in either order-eight tree, this also excludes every ℓ^* -coloring of either spider for $\ell \geq 7$.

Finally, every tree of order less than 6 has a valid 7^* -coloring because every subset is all-6-isolating. Every tree of order 7 has one by assigning a distinct color to each vertex, since deleting a nonempty closed neighborhood leaves at most five vertices. A tree of order 6 is necessarily excluded: any weak partition into seven classes has an empty class, and the empty set is not all-6-isolating. Thus the two spiders above are exactly the minimum-order counterexamples in the admissible class. $\square$

REFERENCES

- [1] G. Boyer, G. C. Farrell, and W. Goddard, *All- k -isolation in trees*, arXiv:2509.11857v1 [math.CO] (2025).